

Ontology, Semantics, Representation, and Interpretation

Toward a Layered Theory of Computational Meaning

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Final determination

Yes. The concepts can be organized into a coherent computational theory, provided the theory is relational rather than a fixed linear pipeline and distinguishes formal semantics from observer interpretation.

Abstract

Ontology, semantics, representation, information, interpretation, and behavior are often arranged as if they formed a simple stack. Cross-disciplinary analysis shows that the distinctions are real, but the stack is too strong. Philosophical ontology studies what exists and how categories of being are related; computational ontologies specify a vocabulary and constraints for a domain. Formal semantics assigns denotations, truth conditions, transitions, or observable behavior to well-formed expressions. Information measures or characterizes distinctions, correlations, descriptions, or truth-bearing content under different theories. Representation is a vehicle-target relation that selectively preserves and exposes structure. Interpretation is the situated application of rules, conventions, prior knowledge, and goals by a system or community. Behavior is a causal outcome produced by an agent-system-environment process and is not entailed by representation or interpretation alone.

The evidence supports a coherent computational theory, but not the proposed one-way sequence Ontology -> Semantic State -> Representation -> Interpretation -> Behavior. Ontology and semantics are mutually constraining specifications; representations may encode syntactic structures, semantic structures, or both; interpretation can establish, recover, revise, or reject semantic assignments; information is typed and may exist without meaning in Shannon's sense; and behavior feeds back into interpretation and model revision. The minimal architecture therefore consists of six roles linked by typed relations: domain commitments, system state, semantic assignment, representational realization, interpretive process, and situated action, with information defined relative to a chosen theory and projection defined as a declared transformation between target and accessible vehicle. The distinctions are largely established separately in philosophy, logic, programming languages, ontology engineering, semiotics, and cognitive science. The warranted contribution is a cross-disciplinary synthesis and diagnostic framework, not a new foundational vocabulary.

Primary contribution: computational meaningfulness should be analyzed as a constrained relation among domain commitments, formal or socially stabilized semantic assignments, representational vehicles, interpreters, and action contexts; no single layer possesses meaning in isolation.

1. Motivation

Computational systems manipulate physical states according to engineered rules. Humans and other systems nevertheless treat some of those states as files, instructions, maps, prices, permissions, identities, or claims. The central theoretical problem is not whether bits can be arranged into layers. It is what licenses movement from physical or formal difference to semantic significance, and which distinctions survive changes of notation, modality, implementation, or observer.

Conflation creates recurrent errors. A database schema is called an ontology although it may encode only local storage constraints. A symbol is said to contain meaning although its denotation depends on a semantic interpretation. A user interface is described as the semantic layer although it is one representation of system state. Successful behavior is treated as proof of shared understanding although equivalent outcomes may arise from different interpretations, heuristics, or accidents. Conversely, strong anti-representational arguments sometimes deny distinctions that formal computer science demonstrably uses to prove compiler correctness, program equivalence, and knowledge-base entailment.

This paper asks one question: Can ontology, semantics, representation, interpretation, and behavior be organized into a coherent computational theory without contradicting existing philosophy and computer science? The test is conservative. A distinction is retained only when established theories assign it a separable explanatory role or when its removal produces a demonstrable loss of explanatory power.

2. Literature Review

2.1 Method

The review is a structured conceptual synthesis of foundational and primary sources from analytic philosophy, formal ontology, knowledge representation, model theory, programming-language semantics, information theory, semiotics, cognitive science, linguistics, distributed cognition, cybernetics, information science, and HCI. Sources were selected when they define one of the focal concepts, specify a formal relation among them, or provide a counterargument to their separation.

Each source was coded for: the relation it presupposes; whether meaning is intrinsic, assigned, causal, inferential, or use-dependent; whether an interpreter is explicit; whether the account is formal or empirical; and what would count as identity or change. Prior project papers were used only as background and consistency constraints, not as evidence for the layered model.

2.2 Foundational tensions

- Realism versus conceptualism: ontology may concern reality itself or an explicit conceptualization used by a community.
- Truth-conditional versus use-based semantics: meaning may be modeled by denotation and truth conditions, inferential role, convention, intention, or practical use.
- Intrinsic versus derived intentionality: computational states may possess only observer-relative meaning, or may acquire content through causal, functional, biological, or social relations.
- Syntactic versus semantic information: Shannon information excludes meaning, while semantic theories add truth, reference, relevance, or viability conditions.
- Internal versus distributed interpretation: interpretation may be assigned to an individual mind, a formal evaluator, an organization, or a coupled cognitive system.
- Linear versus recursive architecture: software diagrams often imply a pipeline, whereas semiotics, learning, and interaction require feedback and revision.

3. Existing Definitions

3.1 Ontology

In philosophical usage, ontology concerns what exists and the most general categories and relations of being. It asks what kinds of entities there are, what identity conditions they possess, and how dependence, parthood, persistence, possibility, and modality are structured. This is not a data model and not a vocabulary file.

In AI and information systems, an ontology is an explicit specification of a conceptualization, later refined as a formal specification of a shared conceptualization. Gruber's account emphasizes representational vocabulary and portability; Guarino distinguishes a conceptualization from the logical theory that approximates its intended models. Computational ontologies commonly contain classes or types, relations, functions or properties, individuals, axioms, constraints, and sometimes processes, temporal relations, or modal commitments. OWL supplies a formal syntax and model-theoretic semantics for such specifications.

A domain ontology is reusable commitment about a domain. A conceptual model is usually designed for a particular application. A schema constrains the organization and validity of stored or exchanged data. A taxonomy primarily orders categories, typically through broader-narrower or subtype relations. These artifacts may overlap, but none is interchangeable with philosophical ontology.

Term	Primary role	Typical contents	What it does not by itself establish
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Philosophical ontology	Account of being and categories	Entities, identity, dependence, modality, relations	A machine-readable vocabulary
Formal ontology	Domain-independent structural theory	Parthood, dependence, universals, particulars, time	A local application schema
Computational/domain ontology	Shared explicit domain commitment	Classes, relations, axioms, constraints, instances	That the modeled categories are metaphysically fundamental
Conceptual model	Application-oriented abstraction	Entities, relationships, processes, rules	Reusable domain consensus
Schema	Data structure and validity constraints	Fields, types, keys, cardinalities	Meaning beyond declared or external semantics
Taxonomy	Classification hierarchy	Categories and broader/narrower relations	Rich relational or modal theory

3.2 Semantics

Semantics is not one relation. In formal logic, an interpretation maps nonlogical vocabulary to a domain and assigns truth conditions to sentences. Tarski's work sharply separates object language and metalanguage and defines truth for formalized languages relative to an interpretation. Model theory generalizes this relation between formal languages and structures.

Programming-language semantics assigns mathematical meaning or execution behavior to programs. Denotational semantics maps expressions to mathematical objects; operational semantics specifies transitions of an abstract machine; axiomatic semantics characterizes programs through assertions and proof rules. These are alternative semantic descriptions of computation, not psychological interpretation by users. Compiler correctness is usually expressed as preservation or refinement between source and target semantics, showing that representation and implementation may change while declared behavior remains equivalent.

Linguistics and philosophy of language introduce truth conditions, reference, compositionality, sense, speaker meaning, convention, context, and pragmatic enrichment. Semiotics treats signification as a triadic or systemic relation. Cognitive and embodied approaches add grounding in perception, action, and environment. Consequently, a general definition must avoid identifying semantics solely with human understanding or solely with formal denotation.

Canonical definition used here: semantics is a rule-governed assignment of content, denotation, truth conditions, inferential role, or behavior to expressions, states, or signs within a specified system and domain. A semantic state is the currently applicable set of content-bearing assignments and relations under that semantics. It is not assumed to be observer-independent; its status depends on whether the semantic theory is formal, social, biological, or psychological.

3.3 Representation

Representation is a selective relation among a vehicle, a target, a mapping or correspondence, and an interpretive practice. A representation may be physical or formal; internal or public; symbolic, iconic, indexical, diagrammatic, linguistic, numerical, or executable. It can encode syntax without preserving target meaning, encode a semantic model, or provide an interface through which semantic consequences become inspectable and actionable.

Representations do not possess semantics in one uniform sense. In a formal system, a representation has assigned semantics by definition of an interpretation function. In social practice, a sign vehicle carries conventional or use-stabilized meaning. In cognition, content may depend on causal, teleological, functional, inferential, or ecological relations. The vehicle alone is insufficient in every account except a strong intrinsic-content theory, for which computational evidence is weak and philosophical controversy remains substantial.

3.4 Interpretation

Interpretation has at least three established senses. In logic, an interpretation is a mathematical assignment from vocabulary to a domain. In semiotics and philosophy of language, interpretation is the production or uptake of significance under conventions, context, and inferential practices. In cognition and HCI, interpretation is an empirically realized process by which an agent forms beliefs, expectations, classifications, or action-relevant understandings from available signs.

These senses must not be collapsed. A model-theoretic interpretation can be completely specified without a conscious interpreter. A human interpretation can diverge from the intended formal semantics. Multiple human or institutional interpretations may coexist for the same vehicle, and one observer may revise interpretation after action or feedback.

Canonical definition used here: interpretation is the application or construction of semantic relations by an interpreting system under a context, competence, history, and purpose. Its result may be a denotation, belief, inference, expectation, classification, or action policy. Interpretation differs from semantics because semantics specifies or stabilizes admissible content relations, whereas interpretation is an event, process, or act that applies, recovers, negotiates, or revises them.

3.5 Information

Shannon information quantifies uncertainty and coding constraints over possible messages; it explicitly brackets meaning. Algorithmic information measures description length relative to a machine. Semantic information theories add well-formedness, meaningfulness, truth, reference, or causal relevance. Ecological information concerns structured invariants available in organism-environment relations. Pragmatic accounts concern information's effect on decision, action, or viability.

No unqualified statement such as 'the information remained the same' is therefore acceptable. The theory, random variables or description language, semantic interpretation, task, observer, and tolerance must be named. A byte-preserving transformation may preserve Shannon information while changing meaning through a different codebook; a lossy representation may reduce recoverable data yet preserve all information relevant to a task.

Canonical definition used here: information is a typed measure or characterization of distinctions, correlations, descriptions, or content relative to a specified source, code, semantic relation, or use. Information is not treated as a synonym for data, meaning, or interpretation.

3.6 Projection

Projection is retained as an analytical role rather than a new substance. It is a declared transformation from a target description or state into an accessible representational form under a contract that identifies preserved, transformed, omitted, and introduced properties, available operations, observer assumptions, intended uses, and failure conditions. The term coordinates existing notions such as abstraction functions, views, encodings, lenses, simulations, and model morphisms. It adds value only when this declaration is explicit.

4. Comparative Analysis

Concept	Question answered	Primary relata	Identity criterion	Characteristic failure
Ontology	What entities and relations are admitted?	Domain, categories, identity and constraints	Same commitments or intended models under declared scope	Category error or incompatible commitment
Semantics	What do expressions or states denote, imply, or do?	Syntax/state, domain/model, assignment rules	Same denotation, truth conditions, transition behavior, or consequences	Undefined, ambiguous, inconsistent, or non-compositional assignment

Information	What distinctions or content are available under a theory?	Source, alternatives, code or semantic relation	Same measure, recoverability, truth-bearing content, or relevance	Loss, noise, falsehood, irrelevance, or theory mismatch
Representation	What vehicle gives selective access to what target?	Vehicle, target, mapping, practice	Typed preservation under a declared equivalence	Omission, distortion, opacity, or unsupported inference
Interpretation	How is significance applied or constructed?	Interpreter, vehicle, context, conventions, goals	Convergent denotation, belief, inference, or policy under scope	Misinterpretation, underdetermination, or disagreement
Behavior	What does the situated system do?	Agent/system, environment, state, action, feedback	Equivalent observable action traces or outcomes under conditions	Performance divergence, adaptation, or confounding

4.1 What is separable?

Ontology and semantics are distinguishable but interdependent. An ontology supplies or constrains the kinds of entities over which a semantics ranges; a semantics makes the ontology's expressions truth-evaluable or inferentially usable. Ontology without semantics may be an uninterpreted signature or informal categorization. Semantics without ontological commitment is impossible in a broad Quinean sense because variables and denotations range over something, but it need not commit to a rich domain ontology.

Semantics and representation are distinguishable because multiple syntactic or material vehicles can receive the same semantic assignment, and one vehicle can receive different assignments under different interpretations. Formal semantics relies on representations to state expressions, models, and rules, but does not identify those tokens with their denotations.

Semantics and interpretation partially overlap in ordinary language but separate in formal and empirical analysis. Formal semantics can be stipulated independently of any person's success in interpreting it. Human interpretation can be studied as a process that approximates, contests, or ignores intended semantics.

Behavior is downstream only in a weak causal sense. Semantic and representational structures constrain possible behavior, but action also depends on goals, memory, competence, attention, incentives, embodiment, environmental state, and stochasticity. Behavior supplies evidence about interpretation but never uniquely identifies it without additional assumptions.

5. Candidate Layered Model

5.1 Rejection of the simple chain

The candidate chain Ontology → Semantic State → Representation → Interpretation → Behavior is rejected as a universal architecture for four reasons. First, ontologies are themselves representations with semantics. Second, representations can precede or induce semantic stabilization rather than merely render a prior semantic state. Third, interpretation may modify the effective ontology and semantic state. Fourth, behavior feeds back through observation, learning, and institutional revision.

5.2 Minimal relational architecture

The minimum supported architecture contains six roles and seven typed relations. Roles are not implementation layers; one artifact may realize several roles, and one role may be distributed across artifacts and agents.

Role	Formal symbol	Definition
Domain commitments	O	Admissible entity, relation, identity, modality, and constraint commitments for the inquiry or system.

System state	X	Physical, computational, social, or modeled condition at a time.
Semantic assignment	S	Rule-governed assignment of denotation, content, truth conditions, inferential consequences, or transition behavior.
Representational realization	R	Vehicle or structure providing selective access to a target under a mapping.
Interpretive process	I	Contextual application, recovery, negotiation, or revision of semantic relations by a system or practice.
Situated behavior	B	Observable action or state transition produced in an agent-system-environment context.

Core relations:

- Commitment: O constrains the admissible domains and identity conditions of S.
- Realization: a projection P maps selected aspects of X or S into R under an explicit preservation contract.
- Interpretation: I applies an interpretation function or learned practice to R in context C, yielding an interpreted state J.
- Affordance and policy mediation: J, goals G, capabilities K, and environment E jointly constrain an action policy π_i .
- Behavior: B is sampled or produced from π_i under current conditions; it is not a direct function of R alone.
- Evidence: observations of B and X update beliefs about I, S, or O but underdetermine them without intervention or auxiliary assumptions.
- Feedback: interpretation and behavior may revise O, S, P, conventions, and future state.

A compact expression is:

$$R = P(X, S \mid O, C_p); \quad J = I(R \mid C, H, K); \quad B \sim \pi(J, G, E, K); \quad \text{observations update } \{O, S, P, I, \pi\}$$

This formalization is intentionally schematic. It blocks three invalid inferences: same representation implies same interpretation; same information implies same semantics; and same behavior implies same understanding.

6. Cross-Disciplinary Synthesis

Discipline	Ontology-equivalent	Semantics-equivalent	Representation-equivalent	Interpretation-equivalent	Behavior/evidence
Analytic philosophy	Ontological commitments; categories	Reference, truth conditions, sense, use	Signs, sentences, models	Speaker/hearer uptake; interpretive charity	Judgments, assertions, inferential practice
Formal logic/model theory	Domain and signature	Satisfaction and interpretation functions	Formulas, structures, proofs	Mathematical assignment, not psychological uptake	Entailment and proof consequences
Programming languages	Types, values, stores, events	Denotational, operational, axiomatic semantics	Source, AST, IR, bytecode	Evaluator/compiler and programmer understanding	Execution traces, outputs, safety properties
AI/knowledge representation	Domain ontology and predicates	Model-theoretic or procedural meaning	Knowledge base, graph, embeddings	Reasoner or agent use; human reading	Entailments, decisions, plans
Information science	Entities, concepts, document kinds	Aboutness, indexing meaning, relevance	Metadata, records, classifications	Indexer and seeker interpretation	Retrieval, selection, use

Linguistics	Discourse entities and event structures	Lexical, compositional, dynamic semantics	Utterances, syntax, gesture	Pragmatic enrichment and discourse update	Responses, inferences, coordination
Cognitive science	World and task models	Mental content, inferential or functional role	Internal/external structures	Perceptual and cognitive construction	Choice, recall, learning, action
HCI	Task and system entities	Intended system and domain meaning	Interface states and interaction forms	User sense-making and system inference	Interaction traces and outcomes

7. What Actually Changes?

Transformation	Ontology	Semantics	Information	Representation	Interpretation	Behavior
Notation change	Usually invariant if symbols are systematically renamed	Invariant only under semantics-preserving translation	May preserve formal content; description length changes	Changes	May change effort or error	May change speed/error despite semantic equivalence
Visualization change	Usually invariant	Declared relations may be preserved selectively	Recoverability and salience can change	Changes substantially	Inference and attention may change	Comparison, search, and error behavior often change
Natural-language change	Target commitments may remain; lexical categories may not align	Translation may preserve propositions imperfectly	Semantic and pragmatic content may shift	Changes	Culture and competence alter uptake	Coordination and decision behavior may change
Modality change	Can remain invariant	Can remain invariant under scoped content mapping	Channel capacity and recoverability differ	Changes	Perceptual processing differs	Latency, workload, access, and error differ
Interaction technique change	System domain may remain	Operation denotations may remain	Available state evidence can differ	Interactive representation changes	Action expectations change	Feasible and selected action sequences change
Rendering technology change	Normally invariant	Normally invariant if rendering is faithful	Signal properties may change	Material realization changes	May change under perceptual constraints	Can change through latency, fidelity, or accessibility
Implementation change	Can remain invariant	Behavioral semantics must be proved preserved, not assumed	Internal descriptions and traces change	Internal representations change	Human interpretation may be unaffected	Observable behavior invariant only under specified equivalence

8. Comparative Case Studies

8.1 Mathematical theorem

Target: a theorem and its proof-theoretic or model-theoretic consequences. Ontology: mathematical objects, relations, and proof objects admitted by the foundational system. Semantic structure: satisfaction, entailment, and validity. Representations: symbolic notation, geometric diagram, prose proof, and executable proof assistant term. A notation change can preserve formal semantics while changing cognitive accessibility. A diagram may expose

invariants and suppress exceptional cases. A proof assistant representation adds machine-checkable dependencies and removes some pragmatic ambiguity. Equivalent theorem statements do not imply equivalent proof understanding.

8.2 Transportation network

Target: locations, connections, schedules, capacities, and routing constraints. A geographic map preserves spatial geometry; a subway diagram preserves connectivity and ordering while distorting distance; an adjacency matrix preserves edge incidence and supports algebraic operations; a routing graph adds weights and operational constraints. The ontology can remain approximately stable while semantic relations and available information differ by scope. Interpretation depends on the user's task: navigation, network planning, accessibility assessment, or capacity optimization. Behavioral route choice can change without any change to the underlying network.

8.3 Computational workflow

Target: a computation and its permitted state transitions. Source code, abstract syntax tree, bytecode, execution graph, and user interface are not simply visual variants. They expose different entities and operations. A compiler seeks semantic preservation between source and target under a declared observation relation. The AST preserves syntactic composition; bytecode preserves executable behavior but can obscure source abstractions; an execution graph exposes causal or control dependencies; a UI exposes selected domain actions rather than program operations. Human interpretation and behavior vary by representation even where compiler-level observational equivalence holds.

9. Counterarguments and Falsification

Claim under test	Contrary evidence or argument	Consequence
Ontology and semantics are separable	Every semantics quantifies over or denotes some domain; intended models partly constitute ontology	Treat them as distinguishable roles with mutual constraints, not independent layers
Semantics is independent of interpretation	Natural-language and social meaning depend on use, context, and communal practices	Limit independence claim to stipulated formal semantics; model social semantics as stabilized interpretive practice
Representation merely exposes prior semantics	Diagrams, notation, and interaction can restructure categories, inference, and later semantic commitments	Allow representation and interpretation to revise effective semantic state and ontology
Information is a neutral middle layer	Shannon information excludes meaning; semantic information theories disagree on truth and relevance	Type every information claim; do not place one generic information layer in the architecture
Behavior reveals interpretation	Behavior is multiply realizable and confounded by skill, goals, incentives, environment, and chance	Require interventions, process measures, and competing-model tests
Software architecture already proves the model	Layered architectures are engineering decompositions, often leaky and application-specific	Use architecture as evidence of practical separability, not metaphysical proof
Computers possess intrinsic semantics	Searle and derived-intentionality arguments deny that syntax alone suffices; causal and functional theories offer alternatives	Remain neutral on intrinsic machine understanding; distinguish formal semantics from phenomenal or intentional meaning

10. Threats to Validity

Terminological heterogeneity is the principal threat. 'Interpretation' names a mathematical assignment in logic and a cognitive or cultural process elsewhere. 'Ontology' names both reality-directed inquiry and engineered artifacts. The synthesis controls this by defining roles rather than asserting lexical identity.

Formal and empirical results operate at different levels. A semantics-preservation theorem establishes equivalence under stipulated observations; it does not establish equal user understanding. Behavioral studies show effects under particular tasks and populations; they do not establish universal semantic mechanisms.

The review is cross-disciplinary rather than exhaustive. Hermeneutics, phenomenology, enactivism, and social theory contain richer accounts of interpretation than can be formalized here. Their omission would bias the framework toward formalism, so the model explicitly permits socially stabilized and enacted semantics while refusing to treat them as computationally complete.

The architecture may be too general to predict behavior without domain-specific mechanisms. This is intentional: it is a conceptual bridge and diagnostic framework, not a complete cognitive theory. Its empirical value depends on whether it supports clearer hypotheses, transformation analyses, and falsification tests than undifferentiated language does.

11. Discussion

The distinctions are not inventions. Philosophical ontology, model-theoretic semantics, semiotic representation, and interpretive practice have long been separated in their home disciplines. Computer science adds unusually sharp demonstrations: a program's syntax, denotation, operational behavior, intermediate representation, compiled implementation, and user-facing presentation can be related by proofs without being identical. Ontology engineering further separates a domain commitment from a representation language and from application data.

What is missing is not another layer but a shared account of how the roles compose. The proposed relational architecture supplies that account conservatively. Meaningfulness is not located in the ontology, semantic state, representation, interpreter, or behavior alone. It arises when representational vehicles are governed by semantic relations grounded or stabilized within a practice and successfully constrain inference or action concerning a target. Formal systems can instantiate this relation by stipulation and proof; human meaning adds competence, history, convention, embodiment, and purpose.

For later work on semantic invariance, the result is precise. What may remain invariant is never 'meaning' without qualification. Candidate invariants include ontological commitment, denotation, truth conditions, inferential consequences, operational behavior, recoverable task information, interpreted belief, or action outcome. Each requires a declared semantics, context, observer class, admissible transformation, and tolerance. Projection analysis should therefore state which of these is claimed to remain invariant and which may change.

12. Conclusion

Ontology, semantics, representation, interpretation, and behavior can be organized into a coherent computational theory without contradicting existing philosophy and computer science. The evidence does not support a strict hierarchy or a universal one-way pipeline. It supports a relational architecture in which domain commitments constrain semantic assignments; projections realize selected aspects of states and semantics as representations; interpreters apply or construct significance under context; and behavior emerges through goals, capabilities, environment, and feedback.

The distinctions are already substantially established, but under incompatible vocabularies and levels of analysis. A genuinely new primitive is not warranted. A cross-disciplinary synthesis is warranted because it prevents category errors and supplies a common method for analyzing invariance, transformation, epistemic access, and behavior. The central theoretical constraint is: no computational artifact is meaningful solely by virtue of its physical form, and no observed behavior uniquely determines the semantics or interpretation that produced it.

13. Required Artifacts

A. Canonical Definitions

Concept	Canonical definition
Ontology	A set of commitments about admissible entities, categories, relations, identity conditions, modalities, and constraints in a domain; in computation, an explicit formal specification approximating a shared conceptualization.
Semantic state	The content-bearing assignments and relations currently licensed by a specified semantics for a system or discourse state.
Information	A typed measure or characterization of distinctions, correlations, descriptions, or content relative to a specified source, code, semantic relation, or use.
Representation	A materially or formally realized vehicle that provides selective access to a declared target under a mapping and interpretive practice.
Interpretation	The application or construction of semantic relations by an interpreting system under context, competence, history, and purpose.
Projection	A declared transformation from a target description or state to an accessible representational form, specifying preservation, transformation, omission, introduction, operations, assumptions, uses, and failure conditions.

B. Relationship Diagram

The diagram is relational rather than strictly layered:

Constraint / feedback	Domain commitments O	Semantic assignment S	Projection P	Representation R	Interpretation I	Behavior B
Forward role	Defines admissible domain	Assigns content/behavior	Selects and transforms	Exposes vehicle	Constructs/ applies significance	Produces situated action
Primary input	Inquiry and practice	O plus expressions/states	X/S plus use contract	Projection output	R plus context/history	Interpreted state plus goals/capabilities/environment
Feedback received	Revised by anomalies/practice	Revised by interpretation/evidence	Revised by failures	Redesigned after use	Learns from outcomes	Changes environment and future evidence

C. Transformation Matrix

Table 7 is the reusable transformation matrix. Its rule is that invariance must be typed and tested independently for ontology, formal semantics, information, representation, interpretation, and behavior.

D. Cross-Disciplinary Vocabulary

Neutral role term	Philosophy / logic	AI / knowledge representation	Programming languages	Linguistics / semiotics	HCI / cognitive science
Domain commitments	Ontology; ontological commitment	Domain ontology; TBox	Types, values, stores, event domains	Discourse ontology; referents	Task/world model

Semantic assignment	Interpretation; satisfaction	Model theory; entailment	Denotational/ operational semantics	Reference; truth conditions; signification	Intended system meaning; conceptual model
Representational vehicle	Sentence, model, symbol	Knowledge graph, assertions, embedding	Syntax, AST, IR, bytecode	Signifier, utterance, diagram	Display, artifact, interface state
Interpretive process	Understanding; hermeneutic uptake	Reasoning or grounding process	Evaluation (formal) or programmer comprehension	Pragmatic interpretation; semiosis	Sense-making, recognition, mental model formation
Behavioral realization	Action or speech act	Decision, plan, policy	Execution trace and output	Response and coordination	Interaction sequence, choice, performance

E. Open Problems

- Grounding: under what causal, functional, biological, or social conditions does formal semantics become intrinsic or non-derived meaning, if ever?
- Semantic-state identity: which equivalence relation should govern claims that semantic state persists across languages, modalities, or implementations?
- Interpretation measurement: how can competing interpretations be inferred without treating task success as equivalent understanding?
- Ontology revision: how should frameworks represent cases where new representations alter the categories through which the target is understood?
- Distributed interpretation: what is the correct unit when meaning is stabilized across people, artifacts, institutions, and computational agents?
- Plural semantics: how should one system support multiple legitimate semantics or stakeholder interpretations without collapsing into ambiguity?
- Machine-generated representations: how can provenance, intended use, and semantic commitments be recovered when no human authored a stable codebook?
- Behavioral equivalence: what interventions distinguish shared semantic competence from policy imitation or coincidentally equivalent outcomes?

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